Exact k-Dispersion for Points in Convex Position: Delaunay Certificates, a Capped Dynamic Program, and Oriented Tight Pairs

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Abstract

Given n points in convex position in the plane and an integer k, the k-dispersion problem asks for k of the points whose minimum pairwise distance r^* is as large as possible. We study exact algorithms and the combinatorial structure of optimal solutions.

Algorithms. We show that a set in convex position has all pairwise distances at least r if and only if it has a locally Delaunay triangulation whose edges all have length at least r. This certificate gives a short correctness proof for a triangulation dynamic program. Because a dispersion decision only asks whether k points fit, each chord of the program needs to store only $k-2$ apex angles, compared exactly by integer in-circle tests. The resulting decision procedure runs in $O(n^3 \log k)$ time and $O(n^2 k)$ space, and solves k-dispersion in $O(n^3 \log k \log n)$ time, improving the $O(n^3 \log^2 n)$ bound of Tkachenko and Wang (SWAT 2026) by a factor $\log n / \log k$. A farthest-first greedy set and the hull perimeter bound r^* to the window $[g, \min(2g, \mathrm{per}(P)/k)]$.

Structure. Call $\{q, a(q)\}$ a *tight pair* when $a(q)$ is the first point counterclockwise from q at distance at least r; a clockwise tight pair $\{b(q), q\}$ is defined symmetrically. We prove that whenever k points pairwise at distance $\geq r$ exist, some such k-set contains a tight pair of one orientation. The proof combines a swap argument based on the quadrilateral inequality with the fact that in every convex polygon some vertex's nearest neighbour is adjacent to it. With a single prescribed orientation the statement holds for $k = 3$ but fails for $k = 4$. We give an explicit 8-point instance whose unique optimal set has no counterclockwise tight pair. It refutes a conjecture that survived 1,306,510 exact randomised tests, and with it every search that grows solutions only from such pairs. The oriented theorem yields an $O(n^2 \log n)$ algorithm for $k = 4$, improving on the previous $O(n^2 \log^2 n)$. We also show that the arc characterisation behind the near-linear algorithm for $k = 3$ fails for every $k \geq 4$, and that natural alternating greedy rules are incomplete.

All algorithms are implemented with exact integer arithmetic and are cross-checked against exhaustive search and an independent solver on a dataset built from convex hulls of random point clouds.

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1 Introduction

Problem. Let P be a set of n points in the plane in *convex position*, i.e. every point of P is a vertex of the convex hull $\mathrm{conv}(P)$. For $2 \le k \le n$, the *k-dispersion* problem (also called max-min dispersion or the maximally separated set problem) asks for $S \subseteq P$, $|S| = k$, maximising $\delta(S) = \min_{p \neq q \in S} |pq|$. We write $r^* = r_k^*(P)$. For general planar point sets the problem is NP-hard [11]; the best exact algorithms run in $n^{O(\sqrt{k})}$ [1] or $O(n^{k-1} \log n)$ time [3]. For points in convex position it is polynomial. Dispersion models facility placement in which facilities must be far apart, and convex position arises when candidate sites lie on the boundary of a convex region.

Known results for convex position. Table 1 summarises the exact algorithms known to us (October 2026).

Table 1: Exact algorithms for k-dispersion of n points in convex position.

Reference	Time	Remarks
Singireddy, Basappa, Mitchell [8]	$O(2^k n^2 \log^2 n)$	bounded search tree, FPT in k
Singireddy, Basappa, Mitchell [8]	$O(n^4 k^2)$	first polynomial algorithm
Tkachenko, Wang [9]	$O(n^{7/2} \log n)$; $O(n^{37/11} \log n)$ exp.	cuttings; randomised
Tkachenko, Wang [10]	$O(n^3 \log^2 n)$	via $O(n^3 \log n)$ max. independent set
Kobayashi et al. [6]	$O(n^2)$	$k = 3$
Tkachenko, Wang [9]	$O(n \log^2 n)$; $O(n \log n)$ exp.	$k = 3$
This paper, Thm. 12	$O(n^3 \log k \log n)$	decision $O(n^3 \log k)$
This paper, Thm. 19	$O(n^2 \log n)$	$k = 4$; decision $O(n^2)$

Contributions.

1. **Delaunay certificate** (Theorem 4): a set in convex position is r-far (all pairwise distances $\ge r$) iff it has a triangulation in which every diagonal is locally Delaunay and every edge has length $\ge r$.

2. **Capped Delaunay DP** (Section 4): the DP value of a chord, as a function of the angle constraint imposed by its parent, is a step function determined by $k-2$ apex points (Lemma 8). This gives an exact decision procedure in $O(n^2 k + T_r \log k) \subseteq O(n^3 \log k)$ time and $O(n^2 k)$ space using only integer orientation and in-circle tests, and an $O(n^3 \log k \log n)$ algorithm (Theorem 12).

3. **Search window** (Lemma 11): $g \le r^* \le \min\{2g, \mathrm{per}(P)/k\}$.

4. **Oriented tight pairs** (Section 6): every feasible instance has a solution containing a tight pair of some orientation (Theorem 15). With one fixed orientation this holds for $k = 3$ (Theorem 17) and fails for $k = 4$ (Proposition 18). As a consequence, $k = 4$ can be solved in $O(n^2 \log n)$ time (Theorem 19).

5. **Limits of greedy structure** (Section 7): the arc characterisation used for $k = 3$ [6, 9] fails for $k \geq 4$, alternating greedy rules are incomplete, and searches seeded only by counterclockwise tight pairs are incorrect. Every claim comes with an explicit counterexample, verified exhaustively in exact arithmetic.

6. **Implementation and experiments** (Section 8), including a *certified* variant that runs an $O(n^2 k)$ greedy and confirms its value with one exact decision.

Positioning. The bound $O(n^3 \log k \log n)$ improves $O(n^3 \log^2 n)$ [10] by a factor $\log n / \log k$ and beats $O(2^k n^2 \log^2 n)$ [8] once $2^k \gtrsim n / \log n$. It is the best known deterministic bound roughly for $\log_2 n \lesssim k \leq n^{o(1)}$. For $k = 4$, Theorem 19 improves the previous best $O(n^2 \log^2 n)$ by a $\log n$ factor. Our DP subproblems are those of [9] in substance. What is new is the certificate-based proof, the k-capped representation, the exact implementation, and the structural results. We could not access the full texts of [10, 8] and compare with their stated bounds only. Section 7 shows that a seed-and-grow search whose seeds are only counterclockwise tight pairs is incorrect for $k = 4$. We do not know whether any published algorithm relies on that property, and we offer the instance of Proposition 18 as a test case.

2 Preliminaries

Table 2: Notation.

$P = (p_0, \ldots, p_{n-1})$	input points in strictly convex position, counterclockwise (ccw); indices mod n
$\lvert pq \rvert$	Euclidean distance; all comparisons use squared distances (integers)
$P(i, j)$	the points strictly between p_i and p_j going ccw from p_i
r-far set	a set whose pairwise distances are all $\geq r$
$\delta(S)$, r^*	minimum pairwise distance of S; $r^* = \max_{\lvert S \rvert = k} \delta(S)$
$a(q)$, $b(q)$	first point ccw (resp. clockwise) from q at distance $\geq r$
succ_S, pred_S	next/previous point of S in ccw order
$\angle abc$	angle at b in triangle abc, in $(0, \pi)$
T_r	number of ccw triples (i, l, j) that are pairwise $\geq r$
$\mathrm{per}(P)$	perimeter of $\mathrm{conv}(P)$

A *triangulation* of a set S in convex position is a maximal set of non-crossing segments between points of S. Its edges are the sides of $\mathrm{conv}(S)$ and the *diagonals*. A diagonal ab with incident triangles abc and bad is *locally Delaunay* if d is not strictly inside the circumcircle of abc, equivalently $\angle acb + \angle adb \leq \pi$. A triangulation is *Delaunay* if no point of S is strictly inside the circumcircle of any triangle.

Exact predicates. For integer coordinates of absolute value at most 2^{29}, the orientation and in-circle determinants fit in signed 128-bit integers. All angle comparisons below reduce to them. For x, y on the same side of the line ij: $\angle ixj > \angle iyj$ iff x is strictly inside the circle through i, y, j. For a convex quadrilateral i, x, l, j: $\angle ixl + \angle ijl > \pi$ iff x is strictly inside the circle through i, l, j.

Lemma 1 (quadrilateral inequality)**.** *If a, b, c, d are in strictly convex position in this cyclic order, then $\lvert ac \rvert + \lvert bd \rvert > \lvert ab \rvert + \lvert cd \rvert$ and $\lvert ac \rvert + \lvert bd \rvert > \lvert ad \rvert + \lvert bc \rvert$.*

Proof. The diagonals ac and bd cross at a point o interior to both. By the strict triangle inequality (o lies on neither segment ab nor cd), $|ao| + |ob| > |ab|$ and $|co| + |od| > |cd|$. Adding gives $|ac| + |bd| > |ab| + |cd|$. The second inequality is symmetric. □

3 A Delaunay certificate for r-far sets

Lemma 2 (Delaunay lemma [4, 5, 2])**.** *If every diagonal of a triangulation T of a finite point set S is locally Delaunay, then T is a Delaunay triangulation of S.*

Lemma 3. *Let T be a Delaunay triangulation of a set S in strictly convex position, $|S| \geq 3$. Then $\delta(S)$ is attained by an edge of T.*

Proof. Let p, q be a closest pair and suppose $pq \notin T$. The triangles of T at p partition the interior angle of $\mathrm{conv}(S)$ at p into wedges. The direction of q from p lies strictly inside the wedge of some triangle puv (not along an edge, by strict convexity), and pq crosses uv. Let C be the circumcircle of puv and A its arc from u to v not containing p. Chords from p inside the wedge end on A. Since q is not strictly inside C, the segment pq leaves the closed disk at some $x \in A$, so $|pq| \geq |px|$. Along A the distance $|px| = 2R\sin(\theta/2)$ (with θ the central angle) first increases and then decreases, so $|px| \geq \min\{|pu|, |pv|\}$. Hence an edge of T is at most as long as the closest pair. □

Theorem 4 (Delaunay certificate)**.** *Let S be in strictly convex position, $|S| \geq 3$, $r > 0$. Then S is r-far iff S has a triangulation T with (a) every edge of length $\geq r$ and (b) every diagonal locally Delaunay.*

Proof. ($\Leftarrow$) By Lemma 2 T is Delaunay, and by Lemma 3 $\delta(S)$ is the length of some edge of T, which is $\geq r$. ($\Rightarrow$) A Delaunay triangulation of S exists and satisfies (b); (a) holds because S is r-far. □

Conditions (a) and (b) are local, which is what makes a dynamic program possible. The algorithm of [9] rests on the same structure, but its correctness proof uses a seven-term distance inequality (their Lemma 14).

4 The capped Delaunay dynamic program

Fix r. A chord (i, j) is *admissible* if $|p_i p_j| \geq r$; a point $l \in P(i, j)$ is an *admissible apex* if $|p_i p_l|, |p_l p_j| \geq r$.

Definition 5. For an admissible chord (i, j), an (i, j)-*configuration* is a pair (X, T) with $X \subseteq P(i, j)$ and T a triangulation of $X \cup \{p_i, p_j\}$ satisfying (a) and (b) of Theorem 4. If $X \neq \emptyset$, its *apex* is the third vertex l of the triangle of T on $p_i p_j$. For $\varphi \in (0, \pi]$ let $F(i, j \mid \varphi)$ be the maximum $|X|$ over configurations with $X = \emptyset$ or $\angle p_i l p_j \leq \varphi$.

If $p_i p_j$ is later a diagonal whose other triangle is $p_j p_i p_t$, its local Delaunay condition is exactly $\angle p_i l p_j \leq \pi - \angle p_i p_t p_j$; the bound φ encodes this.

Lemma 6 (recurrence)**.** *For every admissible chord (i, j) and $\varphi \in (0, \pi]$:*

$$F(i, j \mid \varphi) = \max\Big(\{0\} \cup \big\{\, 1 + F(i, l \mid \pi - \angle p_i p_j p_l) + F(l, j \mid \pi - \angle p_l p_i p_j) \\ : \; l \text{ admissible apex}, \; \angle p_i l p_j \leq \varphi \big\}\Big).$$

Algorithm 1 CappedDecision(P, k, r)

for $g = 2, \dots, n-1$; each admissible chord (i, j) with $j = i + g$ **do**
  $best[1..K] \leftarrow \bot$
  for each admissible apex $l \in P(i, j)$ **do**
    $q_1 \leftarrow \#\{c : L_c(i, l)$ not strictly inside $\bigcirc(p_i, p_l, p_j)\}$; $q_2 \leftarrow \#\{c : L_c(l, j)$ not strictly inside $\bigcirc(p_i, p_l, p_j)\}$
    $v \leftarrow \min(K, 1 + q_1 + q_2)$; keep in $best[v]$ the apex of smaller angle
  end for
  $L_c(i, j) \leftarrow$ smallest-angle apex among $best[c..K]$, for $c = K, \dots, 1$
end for
return true iff some admissible chord has $L_K \neq \bot$

Proof. ($\geq$) Combine optimal configurations for (i, l) and (l, j) with the triangle $p_i p_l p_j$. The new edges have length $\geq r$. The new diagonals $p_i p_l$ and $p_l p_j$ (when the children are nonempty) are locally Delaunay exactly by the children's bounds. The apex is l and satisfies the bound φ. ($\leq$) Removing the apex triangle of an optimal configuration splits it into configurations for (i, l) and (l, j); condition (b) gives their bounds. □

Theorem 7. *For $k \geq 3$: P has an r-far k-set iff some admissible chord has $F(i, j \mid \pi) \geq k - 2$.*

Proof. ($\Rightarrow$) If $S = \{q_1, \dots, q_k\}$ (ccw) is r-far, a Delaunay triangulation of S yields a (q_1, q_k)-configuration with $|X| = k - 2$ (Theorem 4). ($\Leftarrow$) A configuration with $|X| \geq k - 2$ certifies, by Theorem 4, an r-far set of size $\geq k$. □

Thresholds. For an admissible apex l of (i, j) let $v(l) = 1 + F(i, l \mid \pi - \angle p_i p_j p_l) + F(l, j \mid \pi - \angle p_l p_i p_j)$, and for $c \geq 1$ let $\tau_c(i, j) = \min\{\angle p_i l p_j : v(l) \geq c\}$ (or $+\infty$).

Lemma 8. *$\tau_1 \leq \tau_2 \leq \cdots$ and $F(i, j \mid \varphi) = \max\{c \geq 0 : \tau_c \leq \varphi\}$ (with $\tau_0 = 0$). With $K = k - 2$ and $F_K = \min(F, K)$, the values $\tau_1, \dots, \tau_K$ of every chord are determined by the values F_K of shorter chords.*

Proof. The sets $\{l : v(l) \geq c\}$ shrink as c grows, which gives monotonicity. By Lemma 6, $F(i, j \mid \varphi) \geq c$ iff some l with $\angle p_i l p_j \leq \varphi$ has $v(l) \geq c$, i.e. iff $\tau_c \leq \varphi$. Finally $\min(K, v(l)) = \min(K, 1 + F_K(\cdot) + F_K(\cdot))$, and $\tau_{c \leq K}$ depends on v only through $\min(K, v)$. □

We store $\tau_c(i, j)$ as the apex $L_c(i, j)$ attaining it. The query $F_K(i, l \mid \pi - \angle p_i p_j p_l) \geq c$ becomes "$L_c(i, l)$ is not strictly inside the circle through p_i, p_l, p_j", which is monotone in c and is answered by binary search with $O(\log K)$ in-circle tests.

Theorem 9. *Algorithm 1 decides whether P has an r-far k-set in $O(n^2 k + T_r \log k) \subseteq O(n^3 \log k)$ time and $O(n^2 k)$ space, using only exact integer predicates. A witness is recovered in $O(k \log k)$ time.*

Proof. Correctness follows from Theorem 7 and Lemma 8. The inner loop runs T_r times at $O(\log K)$ cost, and each chord costs $O(K)$ more. A witness is recovered by descending from a chord with $L_K \neq \bot$: with demand c at (i, j), take $l = L_c(i, j)$ and send $c_1 = \min(q_1, c - 1)$ to (i, l) and $c - 1 - c_1$ to (l, j). □

Remark 10. Without capping, a chord's step function can have $\Theta(n)$ steps, giving $O(n^3 \log n)$, the bound of [10]. Capping reduces the logarithm to $\log k$ and the memory to $O(n^2 k)$.

5 Optimisation and the search window

Lemma 11 (search window)**.** *Let G be the farthest-first greedy k-set and $g = \delta(G)$. Then $g \le r^* \le \min\{2g, \mathrm{per}(P)/k\}$.*

Proof. $g \le r^*$ since G is feasible. The insertion distances of the greedy are non-increasing, so every point of P is within g of the first $k-1$ greedy points. Two points of an optimal set then share a nearest greedy point, so $r^* \le 2g$. For $k \ge 3$, the convex hull of an optimal set is a convex k-gon with all sides $\ge r^*$, contained in $\mathrm{conv}(P)$. Perimeter is monotone under inclusion of convex sets, so $kr^* \le \mathrm{per}(P)$. For $k = 2$, r^* is the diameter, which is at most $\mathrm{per}(P)/2$. □

Theorem 12. *k-dispersion of n points in convex position with integer coordinates in $[-2^{29}, 2^{29}]$ is solvable exactly in $O(n^3 \log k \log n)$ time and $O(n^2 k)$ space.*

Proof. Binary search over the sorted distinct squared distances inside the window, using Algorithm 1 as the decision procedure. □

6 Oriented tight pairs

Fix r. For $q \in P$ with some point at distance $\ge r$, recall that $a(q)$ and $b(q)$ are the first such points counterclockwise and clockwise from q. A pair $\{q, a(q)\}$ is a *ccw-tight pair* and $\{b(q), q\}$ a *cw-tight pair*. In an r-far set S, a ccw-tight pair is consecutive: $\mathrm{succ}_S(q) = a(q)$, since every point strictly between q and $a(q)$ is closer than r to q.

6.1 A swap lemma

Lemma 13 (swap)**.** *Let S be r-far and $t \in S$.*

(i) If $s = \mathrm{pred}_S(t)$ is a nearest neighbour of t in S, then $(S \setminus \{t\}) \cup \{a(s)\}$ is an r-far k-set containing the ccw-tight pair $\{s, a(s)\}$.

(ii) If $s' = \mathrm{succ}_S(t)$ is a nearest neighbour of t in S, then $(S \setminus \{t\}) \cup \{b(s')\}$ is an r-far k-set containing the cw-tight pair $\{b(s'), s'\}$.

Proof. (i) Since $|st| \ge r$, $w := a(s)$ lies in $P(s,t) \cup \{t\}$. If $w = t$ there is nothing to do. Otherwise w is strictly between s and t, and for each $u \in S \setminus \{s,t\}$ the points s, w, t, u are in strictly convex position in this cyclic order. Lemma 1 gives

$$|st| + |wu| > |sw| + |tu| \;\ge\; r + |st|,$$

using $|sw| \ge r$ (definition of a) and $|tu| \ge |ts|$ (s is a nearest neighbour of t). Hence $|wu| > r$. Also $|ws| \ge r$, so the new set is r-far. (ii) is the mirror image: $w := b(s')$ lies strictly between t and s' (or equals t). For $u \in S \setminus \{t, s'\}$, the points t, w, s', u are in convex position, and $|ts'| + |wu| > |ws'| + |ut| \ge r + |ts'|$. □

6.2 Nearest neighbours in convex polygons

Lemma 14. *Let Q be a convex polygon with $k \ge 3$ vertices and choose for every vertex v a nearest neighbour $\nu(v)$. Then*

(i) the segments $v\nu(v)$ are pairwise non-crossing, and

(ii) for some vertex v, $\nu(v)$ is one of the two polygon neighbours of v.

Proof. (i) Suppose $v\nu(v)$ and $y\nu(y)$ cross properly. Writing $v' = \nu(v)$ and $y' = \nu(y)$, the four points are in convex position in the cyclic order v, y, v', y' (up to reversal), and vy', yv' are opposite sides. Lemma 1 gives $|vv'| + |yy'| > |vy'| + |yv'|$. But $|vv'| \le |vy'|$ and $|yy'| \le |yv'|$ by the choice of nearest neighbours, a contradiction. (ii) Suppose every $v\nu(v)$ is a diagonal. Each diagonal d leaves at least one vertex strictly on each side; let $m(d) \ge 1$ be the smaller of the two counts. Choose $d = xy$ minimising $m(d)$, let C be the side with $m(d)$ vertices, and pick $v \in C$. If $\nu(v)$ were on the other side of xy, the segment $v\nu(v)$ would cross xy properly, contradicting (i). So both endpoints of the diagonal $v\nu(v)$ lie in $C \cup \{x, y\}$. One of its sides then contains only vertices of $C \setminus \{v\}$, so $m(v\nu(v)) \le m(d) - 1$, contradicting the choice of d. □

6.3 The oriented tight-pair theorem

Theorem 15. *If P has an r-far k-set ($k \ge 2$), then it has one that contains a ccw-tight pair or a cw-tight pair.*

Proof. For $k = 2$ take any q with a point at distance $\ge r$; then $\{q, a(q)\}$ works. For $k \ge 3$ let S be r-far. By Lemma 14(ii), applied to the convex polygon S, some $t \in S$ has a polygon neighbour as its nearest neighbour. Lemma 13(i) or (ii) then produces the required set. □

Corollary 16. *In the DP of Section 4, the top level may be restricted to the $2n$ chords $(a(q), q)$ and $(q, b(q))$.*

6.4 One orientation: true for $k = 3$, false for $k = 4$

Earlier versions of this work conjectured that one orientation always suffices:

Property C_1. *If P has an r-far k-set, then it has one containing a ccw-tight pair.*

Theorem 17. *C_1 holds for $k \le 3$ (and so does its mirror image).*

Proof. $k = 2$ is trivial. Let $S = \{x, y, z\}$ be r-far in ccw order. By Lemma 13(i) it suffices that for one of $(s, t, u) \in \{(x, y, z), (y, z, x), (z, x, y)\}$ we have $|tu| \ge |ts|$. Otherwise $|yz| < |xy|$, $|zx| < |yz|$ and $|xy| < |zx|$, i.e. $|xy| < |zx| < |yz| < |xy|$, which is impossible. □

Proposition 18. *C_1 fails for $k = 4$, even at $r = r^*$. In the 8-point instance of Table 3 (strictly convex, ccw order), the unique optimal 4-set is $S = \{p_0, p_2, p_4, p_6\}$ with $r^{*2} = 555009055037$, attained by the diagonal p_2p_6. At $r = r^*$ we have $a(p_0) = p_1$, $a(p_2) = p_3$, $a(p_4) = p_5$ and $a(p_6) = p_7$, none of which is in S. So no r-far 4-set contains a ccw-tight pair. (Every pair $\{b(q), q\}$ with $q \in S$ lies in S, as Theorem 15 predicts.)*

Table 3: The 8-point counterexample to C_1 for $k = 4$ (integer coordinates). Points with odd index are the "blocking" points $a(p_{2i})$.

i	x	y	i	x	y
0	−778191	−140663	4	813287	311458
1	−36313	−275041	5	72900	437065
2	54127	−289061	6	−28204	451365
3	643112	175939	7	−620451	−15151

The claims of Proposition 18 were verified by exhaustive enumeration in exact integer arithmetic by two independent programs.

How the counterexample was found. Randomised exact tests of C_1 on 14,200+39,922 instances (1,306,510 decisions, including targeted perturbations of every other counterexample in this paper) found no failure. The construction instead came from analysing the swap. The swap at $s = \mathrm{pred}_S(t)$ fails only if $a(s)$, which lies in the *cap* beyond the edge st of $\mathrm{conv}(S)$, is closer than r to some other vertex of S. By Lemma 1 this needs $|tu| < |ts|$ for that vertex. We therefore searched, by hill climbing over quadrilaterals, for a convex S in which every cap contains such a point. We then placed one point in each cap, rounded to integers, and checked the result exhaustively. The relevant configurations have small measure, which explains why random sampling missed them; this is a cautionary example for experimental evidence on such questions.

6.5 An $O(n^2 \log n)$ algorithm for $k = 4$

Theorem 19. *For $k = 4$ the decision problem can be solved in $O(n^2)$ time, and 4-dispersion of n points in convex position in $O(n^2 \log n)$ time.*

Proof. By Theorem 15, an r-far 4-set exists iff for some q and some orientation, with $(x, y) = (q, a(q))$ or $(b(q), q)$, there are two points u, u' in the arc of P opposite to the pair, both at distance $\geq r$ from x and y and with $|uu'| \geq r$. (Given a solution with a ccw-tight pair $\{q, a(q)\}$, its other two points lie in $P(a(q), q)$; symmetrically for cw.) For fixed q and orientation, the set F of points of that arc far from both x and y is computed in $O(n)$ time. F is in convex position in ccw order, so its farthest pair is found in $O(|F|)$ time by rotating calipers [7]. Computing $a(\cdot)$ and $b(\cdot)$ takes $O(n^2)$ time, and $2n$ choices of $(q, \text{orientation})$ give $O(n^2)$ in total. Sorting the candidate distances takes $O(n^2 \log n)$, and the binary search makes $O(\log n)$ decisions. □

Corollary 20. *For $k = 5$, replacing the farthest-pair query by the $O(m \log m)$ decision procedure for three points of [9] (Theorem 5 there) gives an $O(n^2 \log n)$ decision procedure and an $O(n^2 \log^2 n)$ algorithm.*

7 Limits of greedy structure

The arc characterisation. For $k = 3$ there is an $O(n \log n)$ decision procedure [9], based on the characterisation (their Lemma 16, from [6]): with $A(i)$ the ccw arc $P[a(p_i), b(p_i)]$,

$$(G_k) \qquad P \text{ has an } r\text{-far } k\text{-set} \iff \exists i : \ A(i) \text{ contains an } r\text{-far } (k-1)\text{-set}.$$

G_3 is true. G_4 is false: Table 4 lists a 6-point instance with no r-far 4-set in which an arc $A(i)$ contains an r-far triple. Our random tests also found failures of G_5 and G_6.

Seeded greedy search. A seed-and-grow search keeps a ccw and a clockwise frontier and repeatedly adds the first point beyond a frontier that is r-far from all chosen points. We tested the following variants:

V_1/V_3: seed $\{q, a(q), b(q)\}$, alternating sides / all side patterns;

V_5/V_4: seed $\{q, a(q)\}$, alternating / all patterns;

$V_4^{\pm}$: seeds $\{q, a(q)\}$ and $\{b(q), q\}$, all patterns.

Proposition 21. *V_1 and V_3 fail for $k = 4$, and V_5 fails for $k = 6$ (Table 4). V_4 fails for $k = 4$: it implies C_1, which fails by Proposition 18.*

All counterexamples are thin convex polygons. In such polygons, points that are adjacent in space can be far apart in the cyclic order, so a search that commits early on one side can block the other.

Table 4: Counterexamples (points in ccw order, 0-based indices). For G_4 no r-far 4-set exists; in the other rows an r-far k-set exists (one is shown) but the property fails.

Property	n	k	r^2	an r-far k-set
C1	8	4	555009055037	0, 2, 4, 6
`(-778191,-140663), (-36313,-275041), (54127,-289061), (643112,175939), (813287,311458), (72900,437065), (-28204,451365), (-620451,-15151)`				
G4	6	4	320756	none
`(-976,42), (4,-49), (971,-41), (979,4), (964,29), (398,49)`				
V1	8	4	2556513005	0, 3, 4, 7
`(-90482,144), (-45601,-629), (13859,-779), (38501,-677), (97951,32), (57181,535), (4941,792), (-19180,765)`				
V5	10	6	878078401	0, 1, 2, 5, 6, 7
`(-94942,276), (-57395,-1876), (-27771,-2581), (-2908,-2795), (62044,-1720), (87508,-655), (48547,2138), (2056,2796), (-24473,2629), (-70081,1423)`				

Conjecture 22. *$V_4^{\pm}$ is complete: whenever P has an r-far k-set, the search seeded with some oriented tight pair and some side pattern reaches k points.*

The conjecture holds for $k \le 3$ by Theorem 15. It held in all 1,006,005 exact tests (45,944 instances, including perturbations of all counterexamples), and on every instance of Table 4. If true, it gives an $O(2^k n^2 k \log n)$ algorithm: the search tree has at most 2^{k-1} nodes per seed and each node scans $O(n)$ points against $\le k$ chosen ones. Given Proposition 18, we regard this evidence with caution.

A certified algorithm. Seeded greedies are sound: every set they output is r-far. Algorithm CERTIFIED binary-searches the window with the greedy (seeds of both orientations, alternating) as predicate, obtaining $\rho \le r^*$. It then runs one exact decision at the successor of ρ. If that fails, $\rho = r^*$; otherwise the exact binary search continues above ρ. The output is always exact, and the time is $O(n^2 k \log n + n^3 \log k)$ whenever the greedy finds r^*.

8 Experiments

Dataset (random points → convex hull). Instances are generated by sampling N random integer points (coordinates up to 10^6) from a distribution, taking the strictly convex hull, and keeping n of its vertices. Any subset of a set in convex position is in convex position. The distributions are uniform square and Gaussian (small hulls, so $n \le 30$ and $n \le 14$), uniform disk, thin annulus, thin 1:20 ellipse, points on a circle, and *cocircular* points: all 324 lattice points on $x^2 + y^2 = 32045^2$, the most degenerate input for in-circle tests. There are three splits: *verify* ($n \le 14$, 280 instances), *medium* ($n \le 40$, 68) and *scaling* ($n \le 200$, 93).

Correctness. Each instance was solved by the capped DP, the certified algorithm, the Tkachenko–Wang DP (direct implementation), exhaustive search and, on *verify*, an independent FMM-based clique solver. Every returned set was re-checked independently. The $k = 4$ algorithm of Theorem 19 agreed with the capped DP on 3,003 random instances and on all $k = 4$ counterexamples.

Running time. Table 6 and Figure 1 report median in-process solve times on one core. The window kept a median of 6.3% (max 27.4%) of the candidate distances and saved 4.5 exact decisions on average. The certified algorithm was certified by its greedy on 93/93 scaling instances.

Structural tests. Table 7 summarises the exhaustive tests of structural properties.

Table 5: Optimum agrees with exhaustive search and witness valid, per distribution: 280/280 (verify) and 68/68 (medium) instances fully correct.

verify ($n \le 14$)

Distribution	inst.	capped	certified	TW	FMM
`square`	40	40/40	40/40	40/40	40/40
`disk`	40	40/40	40/40	40/40	40/40
`gauss`	40	40/40	40/40	40/40	40/40
`annulus`	40	40/40	40/40	40/40	40/40
`thin`	40	40/40	40/40	40/40	40/40
`circle`	40	40/40	40/40	40/40	40/40
`cocircular`	40	40/40	40/40	40/40	40/40

medium ($20 \le n \le 40$)

Distribution	inst.	capped	certified	TW
`square`	8	8/8	8/8	8/8
`disk`	12	12/12	12/12	12/12
`annulus`	12	12/12	12/12	12/12
`thin`	12	12/12	12/12	12/12
`circle`	12	12/12	12/12	12/12
`cocircular`	12	12/12	12/12	12/12

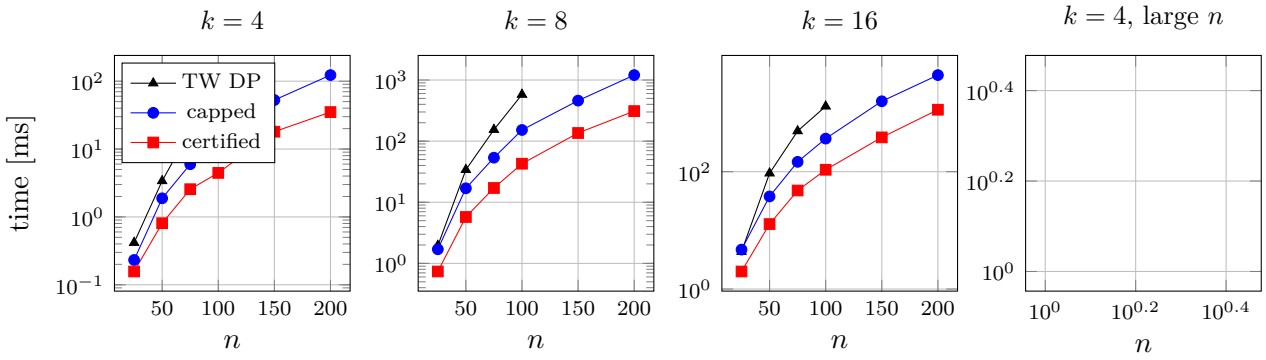


Figure 1: Median solve time vs. n (log scale). The right panel compares the $k = 4$ algorithm of Theorem 19 with the capped DP on random convex polygons, two-chain shapes and circles.

9 Conclusion and open problems

We gave a simple exact $O(n^3 \log k \log n)$ algorithm for k-dispersion in convex position, an $O(n^2 \log n)$ algorithm for $k = 4$, and a structural theory of tight pairs: some optimal set always contains a tight pair of *some* orientation, but not necessarily of a prescribed one once $k \ge 4$. Open questions:

1. Does C_1 fail for every $k \ge 4$? Our lifting experiments ($k = 5$) found no counterexample.
2. Prove or refute Conjecture 22. A proof would give an $O(2^k n^2 k \log n)$ algorithm with a self-contained correctness argument.
3. Can $k = 4$ be solved in $o(n^2)$ time, matching the near-linear bound for $k = 3$? Is there an $o(n^3)$ algorithm for every fixed k?
4. Can the $\log n$ factor of the binary search be removed?

Table 6: Median solve time [ms] and exact decisions on the scaling split (TW only for $n \leq 100$).

k	n	TW	capped	certified	TW/capped	dec. (no window)	dec. (window)	dec. (cert.)
4	25	0.4	0.2	0.2	1.5×	10	6	2
4	50	3.4	1.9	0.8	1.7×	11	8	2
4	75	18.9	6.0	2.6	3.2×	12	9	2
4	100	36.6	16.6	4.5	2.1×	13	10	2
4	150	–	52.7	18.0	–	14	11	2
4	200	–	122.9	35.1	–	15	11	2
8	25	1.9	1.7	0.7	1.2×	9	5	2
8	50	34.0	16.9	5.7	1.9×	11	7	2
8	75	153.9	53.6	17.1	3.0×	12	7	2
8	100	578.0	152.3	42.7	3.8×	14	8	2
8	150	–	460.4	135.8	–	15	8	2
8	200	–	1203.9	310.0	–	15	9	2
16	25	4.3	4.7	2.0	1.0×	9	5	2
16	50	94.1	37.9	12.8	1.7×	11	6	2
16	75	493.4	147.3	47.8	2.6×	12	7	2
16	100	1300.8	366.7	107.9	3.4×	13	7	2
16	150	–	1587.0	384.9	–	14	8	2
16	200	–	4434.8	1139.0	–	15	10	2

Table 7: Exact tests of structural properties: counts of (P, k, r) tested and failures found by random search. "Status" is the proven status.

Property	k	tests	failures	status
C_1 (random hunts, $n \leq 30$)	3–12	429,742	0	false for $k = 4$ (Prop. 18)
C_1 (perturbations)	3–10	876,768	0	
V_4 (random hunts, $n \leq 26$)	3–12	490,671	0	false for $k = 4$ (Prop. 21)
V_4 (perturbations)	3–10	876,002	0	
V_5 (random hunts, $n \leq 22$)	3–9	201,422	5	false for $k = 6$
$V_4^{\pm}$ (hunts and perturbations)	3–10	1,006,005	0	open (Conj. 22)

Reproducibility. Code, data and scripts are in `convex/`: the solver (`convex_dispersion.cpp`), dataset generation, the cross-verification harness, structural experiments, counterexample search, minimisation and independent re-verification (`experiments/check_counterexamples.py`).

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